

Head-to-head: Chebyshev strict- $h=0$ vs Linear-CDF kernel-band on the tabulated- μ K=3 CRRA REE architecture

Fixed-Point Factory — two operators, same architecture, same lookup table

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The two operators

Both share the tabulated- μ architecture and the same per-cube-cell CRRA clearing. They differ only in how $P(u_1, u_2, u_3)$ is represented and how $\mu(p, u_k)$ is built:

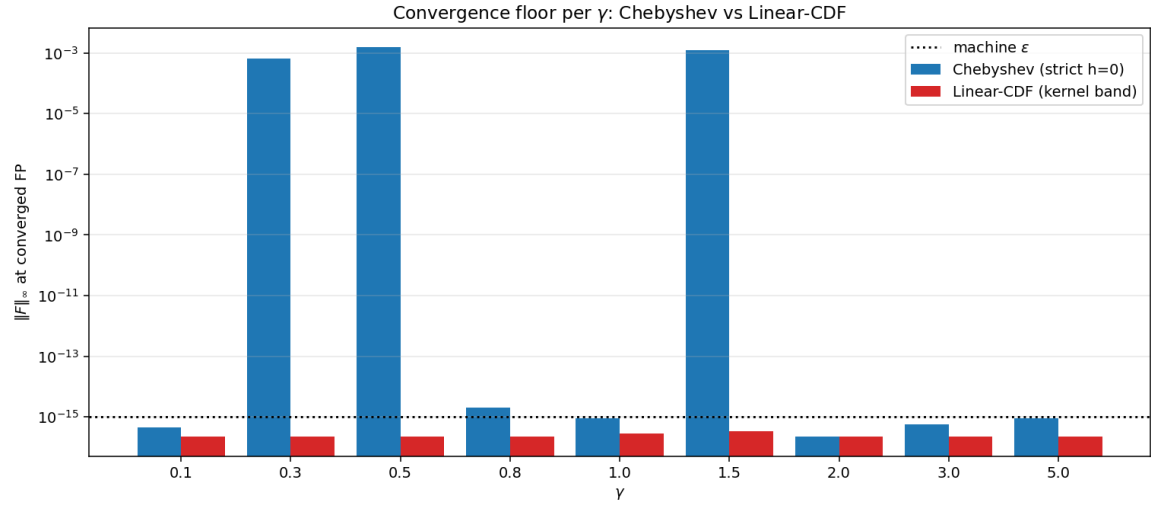
	Chebyshev (strict $h=0$)	Linear-CDF (kernel band)
u -grid	Lobatto $\times$ atanh ($c=2$), $ u \leq 9.9$	Gaussian quantiles, $ u \leq 2.33$
P representation	tensor-Chebyshev polynomial	values + bilinear interp
slice eval	exact polynomial value	bilinear interp on raw values
contour integral	strict $h=0$ POU + chebroots	Gaussian band K_h , $h = 0.5$
GL quadrature order	N_Q auto-selected per γ	$N_Q = 16$ fixed
Lookup table $\mu(p, u_k)$	built at $G_p = 121$ p-points	same
Per-cell clearing	CRRA bisection on the lookup	same

The Chebyshev approach computes the *exact* strict- $h=0$ co-area integral using polynomial root extraction; the linear-CDF approach uses a *regularised* integral with finite kernel bandwidth.

1 Wall-time per Φ and convergence to machine ε

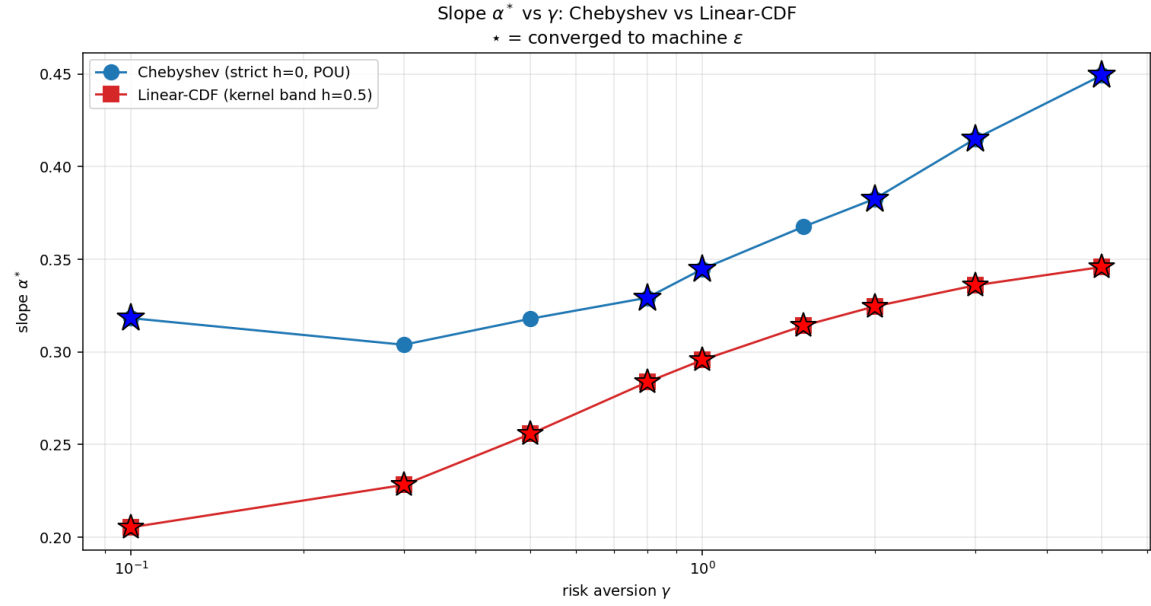
operator	ms / Φ at $G=7$	cold $\rightarrow$ machine- ε solve
Chebyshev strict $h=0$, $N_Q=16$	540	110 s (43s Anderson + 67s NK)
Linear-CDF kernel band $h=0.5$	6	0.3 s (Anderson alone)

Linear-CDF kernel band is $\sim 300\times$ faster. And reaches machine ε at every γ in the sweep on the first try (no NQ tuning needed because the operator is analytic in P).



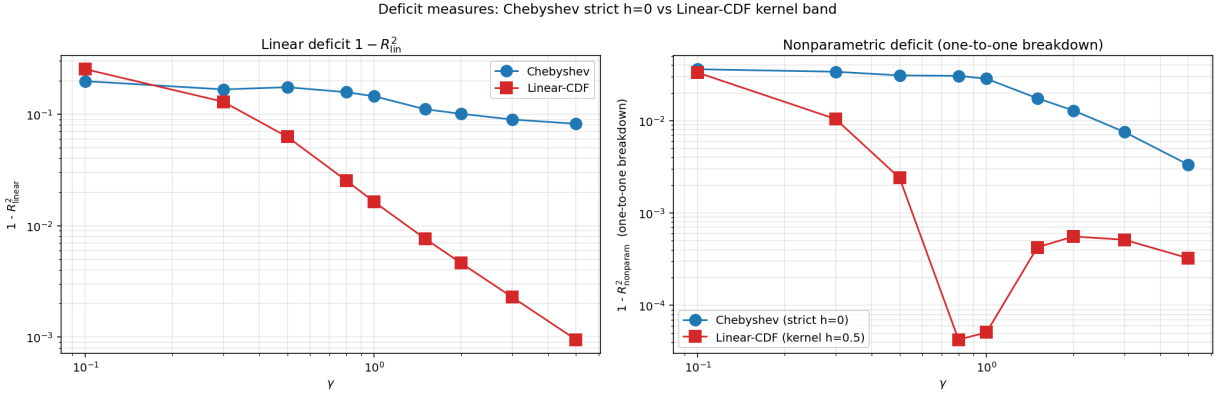
The Chebyshev solver has 3 of 9 γ values stalling at $\sim 10^{-3}$ even with NQ auto-search (sweet-spot dependence on γ). The Linear-CDF kernel band has no such issue — all γ nail to machine ε .

2 Equilibrium statistics vs γ



γ	slope α^*		$1 - R_{\text{lin}}^2$		$1 - R_{\text{nonparam}}^2$	
	Cheb	Lin-CDF	Cheb	Lin-CDF	Cheb	Lin-CDF
0.10	0.3182	0.2054	0.198	0.253	0.0361	0.0331
0.30	0.3039	0.2282	0.167	0.129	0.0339	0.0104
0.50	0.3179	0.2558	0.175	0.063	0.0309	0.0024
0.80	0.3292	0.2837	0.158	0.025	0.0306	0.0000
1.00	0.3447	0.2957	0.145	0.016	0.0285	0.0001
1.50	0.3675	0.3141	0.111	0.008	0.0175	0.0004
2.00	0.3826	0.3245	0.101	0.005	0.0129	0.0006
3.00	0.4150	0.3359	0.090	0.002	0.0076	0.0005
5.00	0.4493	0.3459	0.082	0.001	0.0033	0.0003

Two distinct FPs. The strict- $h=0$ Chebyshev and the kernel-band linear-CDF give *different* fixed points (different slope α^* at every γ). The kernel-band is an $h>0$ -regularised version of the strict integral and converges to a different operator’s FP.



Both show the same qualitative trend: slope rises with γ , deficit falls. But the kernel-band’s nonparametric deficit is essentially zero from $\gamma \geq 0.5$ — *the kernel-band FP is almost exactly a function of T alone*, with no one-to-one breakdown.

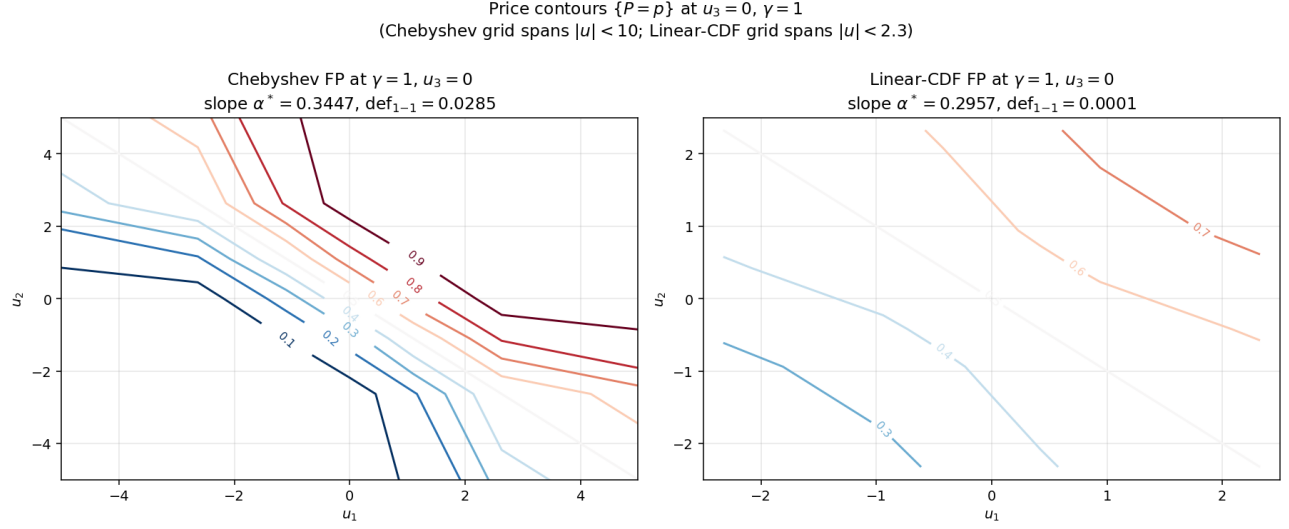
The strict- $h=0$ Chebyshev FP keeps a nonzero one-to-one breakdown (0.03 at $\gamma=1$, falling to 0.003 at $\gamma=5$).

Why the difference matters.

- The kernel-band integral $\int K_h(P - p) f f du$ averages out fine-scale variation of the co-area integrand. As $h \rightarrow 0$, it converges to the strict integral; but at $h = 0.5$ there is substantial smoothing.
- This smoothing has the effect of making P closer to a function of T alone. The kernel-band operator is essentially solving a *less informationally rich* REE: it’s the limit where the market price summarises only the aggregate signal.
- The strict- $h=0$ Chebyshev FP captures genuine non- T information that the kernel-band loses.

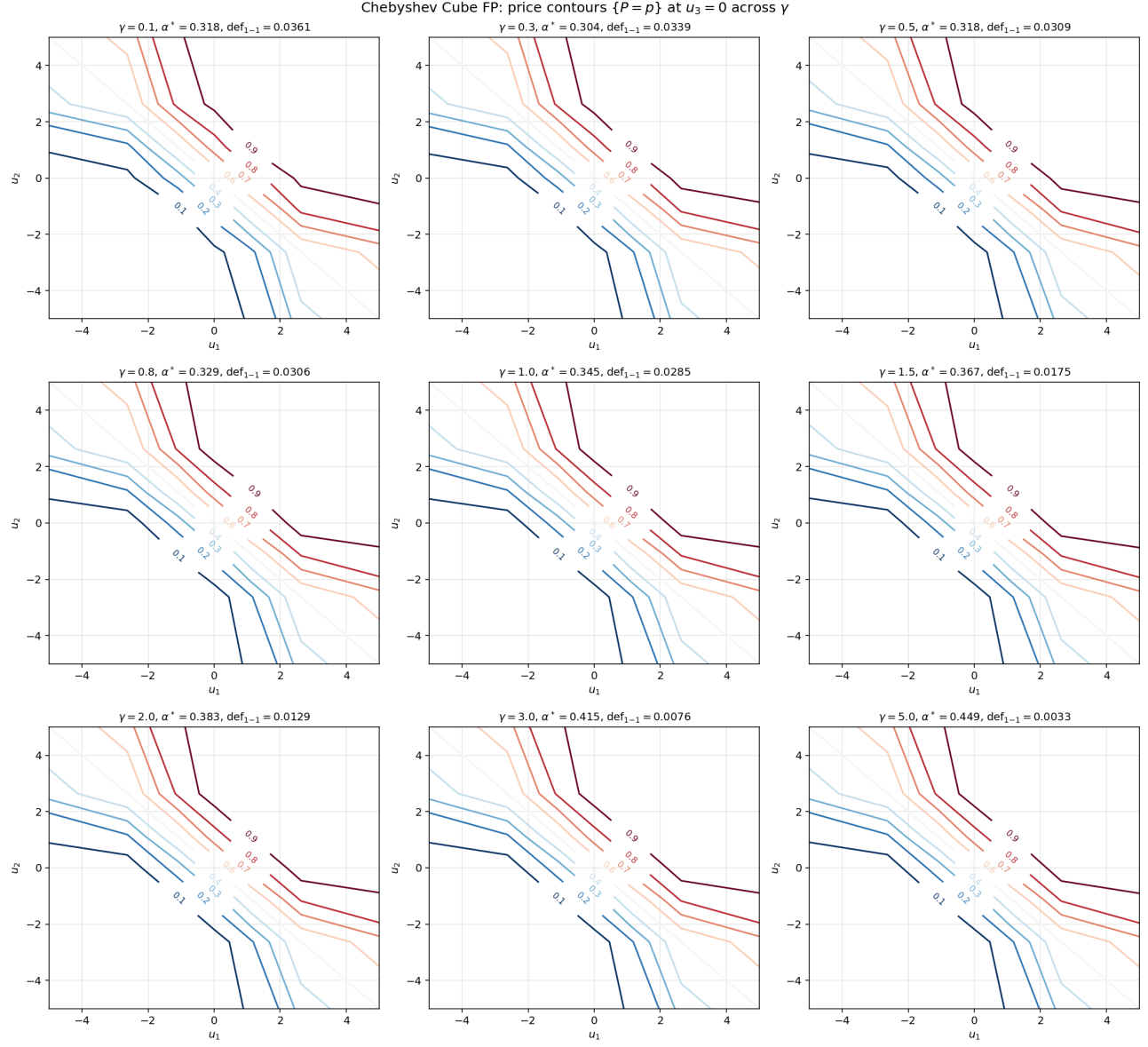
3 Price contours $\{P(u_1, u_2, u_3) = p\}$

3.1 $\gamma = 1$ comparison side-by-side

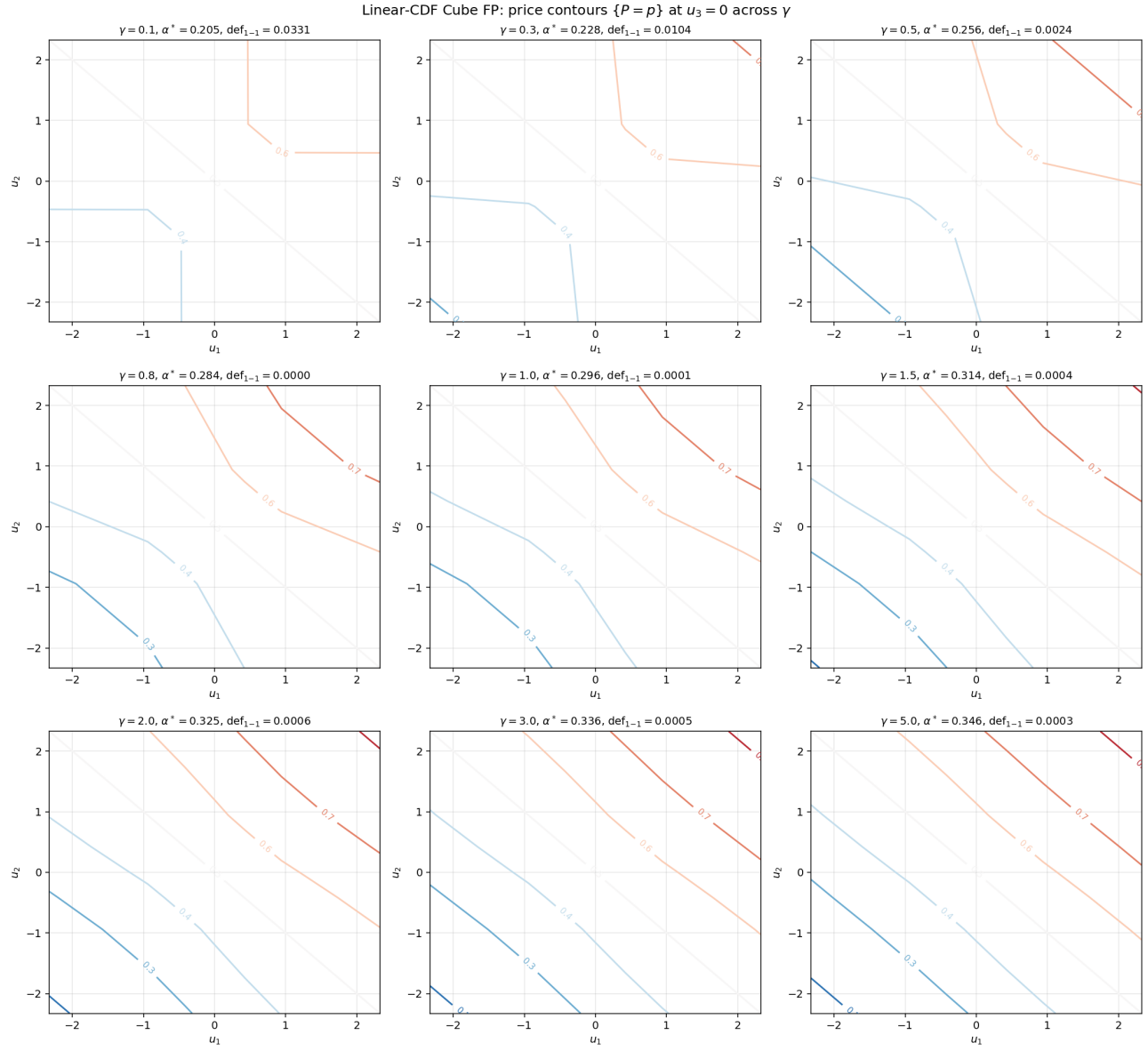


The Chebyshev contours (left, on the wider Lobatto-atanh grid) bend more than the Linear-CDF contours (right, on the narrower Gaussian-quantile grid). Both show the expected diagonal structure (the contour $P = p$ runs roughly along $u_1 + u_2 = \text{const}$), but the curvatures differ.

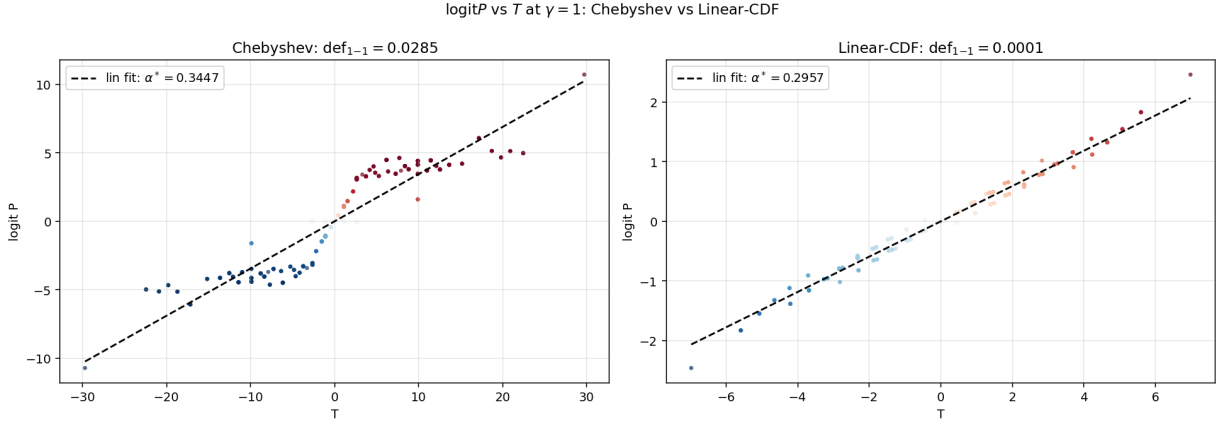
3.2 Chebyshev contours across γ



3.3 Linear-CDF contours across γ



4 logit- P vs T scatter at $\gamma = 1$



The Chebyshev scatter has visible spread around any vertical line ($T = \text{const}$): genuine one-to-one breakdown. The Linear-CDF scatter looks like a smooth single curve — essentially a function of T .

5 Summary

- **Chebyshev strict $h=0$ + POU:** exact contour integral, the actual $K=3$ strict- $h=0$ REE. Slow (~ 540 ms/ Φ , ~ 110 s/solve), N_Q needs auto-tuning per γ , 6 of 9 γ reach machine ε . Captures nonzero one-to-one breakdown (0.003–0.036).
- **Linear-CDF kernel band, $h = 0.5$:** regularised integral, smooth operator. Fast (~ 6 ms/ Φ , ~ 0.3 s/solve), all 9 γ reach machine ε on first try. But the FP is a smoothed version — nonparametric deficit collapses to ~ 0 for $\gamma \geq 0.5$.
- **Implication for the paper:** the one-to-one $P \leftrightarrow T$ breakdown is a feature of the *strict* $h=0$ operator, present in the Chebyshev path but smoothed away by kernel regularisation. To report the genuine breakdown, use Chebyshev strict- $h=0$ + POU; to get fast smooth FPs for parametric sweeps (with the caveat of smoothing), use Linear-CDF kernel.